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From Backlogs to Bots: Generative AI's Impact on Agile Role Evolution

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ABSTRACT

This position paper investigates the transformative impact of generative artificial intelligence (GenAI) on Agile roles. Focusing on the product owner, developer, and scrum master, we analyze how GenAI redefines traditional tasks, encouraging a shift towards more strategic and creative functions. Through practical experience, we illustrate AI's role in enhancing Agile processes, its practices and emphasize the need for Agile practitioners to integrate AI understanding. These results highlight the balance between GenAI's efficiencies and Agile's human-centric principles, proposing research directions for AI-enriched Agile practices that promise to drive innovation and maintain the agility in a technologically progressive era.

1 | Introduction

In the evolving business world, change is not just a constant but the currency of survival and success. Companies are navigating an environment that not only evolves but also accelerates with each technological advance. In this race against time, businesses are forced to adapt at a rapid pace, striving for agility to meet the shifting demands of global markets and consumer expectations [1].

The relentless speed of the business world today is largely fostered by digitalization, a phenomenon that has redefined the essence of operational strategies and organizational structures. As digital technologies weave deeper into business operations, they create new opportunities for growth and innovation while simultaneously disrupting established norms and practices. The current heart of this digital revolution is artificial intelligence (AI) [2]. AI's potential to transform the business landscape promises to revolutionize many workplaces. The integration of AI—especially generative AI—into various business processes is not only optimizing existing operations but is also creating new paradigms for work, employment, and value creation [3].

With this transformative power of AI, a number of jobs will change or evolve, with AI reshaping job profiles and their tasks. The effects are pervasive, within every domain from manufacturing to services witnessing a shift in the skill sets and competencies required to thrive in the new AI workplace [4].

Project management—a discipline central for the operationalization and execution of business strategies—is also confronted with this transformation. The integration of generative AI promises to change project management by providing tools that not only enhance efficiency and accuracy but also elevate decision making and strategic planning. AI's ability to analyze vast datasets in real-time allows for predictive insights, risk assessment, and resource optimization, thereby addressing some of the most challenging aspects of project management.

As this transformation permeates the field, it naturally extends into the specialized domain of Agile project management. Agile, with its emphasis on flexibility, speed, and collaboration, stands to benefit uniquely from AI's capabilities. In Agile settings, AI tools can support refining processes such as sprint planning, backlog management, and team dynamics, enriching the Agile values of customer collaboration and responsiveness to change. The roles of key Agile practitioners, including scrum masters (SMs), product owners (POs), and developers, are thus evolving. They are moving towards a new paradigm where their individual tasks are not just redefined but also enhanced by AI, calling for a renewed understanding of Agile principles in light of AI's potential.

The implications of this shift are vast and complex, warranting a deeper investigation into how AI is redefining Agile project management. This position paper is based on long-term experiences in agile development and aims to explore these implications, offering insights into the synergies and challenges at the intersection of AI and Agile methodologies with the focus on the common roles in an agile setup.

2 | Agile and AI: Background

2.1 | Agile: A Dual Perspective

Agile methodology encompasses two fundamental sides: the cultural and the technical. The cultural aspect of Agile is rooted in values and principles—still strongly influenced from the agile manifesto [5]—that prioritize individuals and their interactions. It focuses on collaboration, customer feedback, and the ability to respond to change—soft skills that are essential for Agile teams to function effectively. These soft skills include communication, adaptability, problem-solving, and a strong team spirit, all of which are crucial for an Agile mindset [6].

On the technical side, Agile is characterized by its frameworks, methods, and underlying practices. This includes various frameworks like e.g. the most common ones, Scrum or Kanban, which provide structure to the Agile process. The "technical" practices which are the smallest building blocks involve routine meetings (such as daily stand-ups, sprint planning, reviews, and retrospectives), artifacts (like product backlogs and sprint backlogs), and roles (such as SM, PO, and team members) [7]. These elements serve as the backbone of Agile processes, ensuring that Agile teams can work in a manner that is iterative, incremental, and continuous.

2.2 | AI: Maximizing Potential Through Skillful Use

AI is a domain of computer science that aims to create systems capable of performing tasks that typically require human intelligence. Thus, it presents a powerful set of tools that have the potential to transform industries and workflows [3, 4]. However, it is clear that just having access to technology like this is not enough. It necessitates a deep understanding of AI's capabilities and limitations, and the skill to apply it in a manner that complements and enhances human effort.

One and probably the most important of the emerging challenges within AI is the field of prompt engineering. This refers to the skillful crafting of inputs and queries to effectively communicate with AI systems, ensuring that the outputs generated are useful, relevant, and aligned with the user's intent. Prompt engineering does not only require technical know-how but also creativity and insight into the workings of AI language models and machine learning algorithms [8]. As AI becomes more prevalent, the ability to interact with and guide AI systems through effective prompts will be a critical skill for maximizing their potential in various applications, including Agile project management.

2.3 | AI and (Agile) Project Management

Mueller et al. [9] shows that there is still an academic interest in understanding AI's impact on management and managerial practices. This shift is particularly noted in project management, where AI applications have evolved from expert systems aiding in decision making to more intricate roles in effort estimation, cost management, and risk management among other areas. The aspect of AI on decision making in project management is analyzed in detail in [10].

An overview based on a survey [9] of 2314 professionals from 129 countries indicates a global acknowledgment of AI's transformative impact on project management. Key areas of influence include data collection and reporting, performance monitoring, and project scheduling. However, the findings also reveal a significant gap in AI training within organizations, with many professionals possessing only a basic level of AI knowledge and experience. Despite that, there is a marked interest among project managers to deepen their understanding of AI, recognizing its potential to revolutionize their field. An interesting example is given by Dam et al. [11] about how AI—not only generative AI—could be used for a better planning and analysis procedure, mainly for the role of the PO.

The survey further underscores the perceived importance of national governments in fostering AI development through investments in technology and education, as well as in legislating and establishing guidelines to ensure ethical and secure AI practices. This collective perspective underscores the critical role of governmental intervention in facilitating the integration of AI within project management and in addressing the broader challenges associated with AI adoption in professional settings.

3 | The Transformation of Agile Roles Through AI

The convergence of Agile and AI began when technologists started recognizing AI's potential to enhance Agile processes. AI's ability to analyze large datasets, automate routine tasks, and provide predictive insights presented opportunities to rethink jobs and tasks to be done in different roles [12]. Thus, the following section is going to discuss AI's role in the current and next evolution of agile roles—more from the viewpoint of technical agility.

Before digging into the role-specific details, it is necessary to mention that all the following aspects perfectly fit to an agile culture that is grounded on experimentation and adaptation. This is the case because using GenAI technologies, especially prompt engineering, needs an experimental culture and the user of such tools needs to adapt [6].

3.1 | The Product Owner: Focus on Creativity and Crafting

The role of the PO is undergoing a significant transformation influenced by the appearance of AI. The traditional focus on product- and backlog management [13] is shifting towards a more creative and strategic direction (*H-PO1*). The PO is now expected to engage more in crafting a product vision and strategy rather than being involved in all the details of backlog items.

The product vision and its strategy rely on the PO's ability to synthesize market trends, user needs, and business goals. While AI can support this process by suggesting new ideas and optimizing existing functionalities, the basic idea as well as the responsibility for the product vision remains with the PO. This ensures that the product aligns with the complex combination of user needs, market demands, and ethical considerations that AI alone cannot navigate. The collaboration between the PO and AI systems necessitates a delicate balance. AI can handle vast amounts of information, but the PO must guide it by providing the correct and relevant information (*H-PO2*). This symbiosis ensures that AI complements the human intelligence of the PO, leading to more innovative and strategically aligned product development.

Furthermore, the classical PO works a lot in the field of requirements engineering—a task traditionally requiring extensive writing of user stories, designing user interfaces, and outlining acceptance criteria. Generative AI has the potential to revolutionize this facet of the PO's work by automating the creation of text, user interfaces, even user journey maps or other requirements-related content [11, 13]. This will allow the PO to allocate more time to high-level strategic tasks mentioned before. AI systems, with its proficiency in managing and analyzing large datasets, could take on the role of the primary backlog item creator and manager. This shift can significantly reduce the administrative burden on the PO, who would then refine the AI-generated backlog to ensure alignment with the product vision.

3.2 | The Dev-Team: From Code to Collaboration

The role of the software developer is being redefined most of all the roles. AI and its capabilities increase the automation of routine tasks (*H-DEV1*), which are especially programming and quality assurance that includes all kind of testing activities. Furthermore, [14] showed that especially in programming, overall AI systems work well without many mistakes. Nevertheless, this automation depends on providing precise inputs reflecting the problem or need—this is where prompt engineering becomes important. It necessitates a deeper collaboration between developers and PO to ensure that AI solutions are furnished with high-quality requirements (*H-DEV2*). These requirements, effectively crafted, serve as sophisticated prompts that enable AI to generate code, thus fostering a more integrated approach with product development.

The consequence of this shift is twofold: On the one hand, it releases developers from the aspect of repetitive coding tasks and engage them in more inventive and cooperative facets of software development. On the other hand, it repositions developers at the leading edge of innovative problem-solving. Their expertise exceeds coding skills: All developers need to become architects of complex software solutions (*H-DEV3*), blending creativity with technical proficiency. This paradigm shift is transformative, signaling a departure from traditional programming roles towards a future where developers get technical innovation leaders. Their role evolves to encompass the conceptualization and execution of advanced software solutions that are responsive to dynamic market needs and technological possibilities.

3.3 | The Scrum Master: New Dimensions of Leadership

The Scrum Master (SM) role is often misunderstood in practical context, sometimes conflated with administrative positions such as a scribe, secretary, or tool administrator for handling JIRA or similar tools. Even if this is a misinterpretation, the rise of AI offers a compelling solution to this: Certain tasks, such as transcription or administrative tool maintenance, can be managed by AI without the need for sophisticated prompting or deep knowledge (*H-SM1*). This allows the SM to fully embody the real key stances: servant leader, facilitator, coach, manager, mentor, teacher, impediment remover, and change agent [15].

AI's capacity to assume routine management tasks creates space for the SM to move towards either a more strategic function from management aspect or focus more on the other stances [15]. Furthermore, AI can support teaching and training through the delivery of educational content either completely by an AI system or in combination with a human trainer (*H-SM2a*). Independent of the concrete training implementation, it should be followed by continuous coaching support by the agile coaching stance (*H-SM2b*). In facilitation, AI can enhance the creativity of meetings such as by generating novel questions for daily stand-ups or extending existing SM tools like the Retromat for retrospectives [16]. To give another example, with the use of generative AI, it is easy and fast to generate topic specific meetings.

The usage of AI complements human expertise and thus is ensuring that the SM can focus on fostering an agile culture with continuous improvement (*H-SM3*). This represents a significant shift in the SM's function, moving towards a more coaching-centric approach, aligning closely with the role of an Agile coach. With this AI-support, the SM can redefine his focus on the less AI-supported stances, particularly in nurturing the team's growth, facilitating empowerment, and driving change, thereby enhancing the agility and effectiveness of the team.

3.4 | The Agile Coach: Towards an AI-Trainer

With the SM going more into a coaching role, this follows the same direction with the Agile coaching role. Similar to this, the coach focuses more on the human aspects than the knowledge and consulting aspects, such as methodologies. Therefore, they focus on nurturing soft skills and interpersonal dynamics that facilitate high-functioning teams (*H-AC1*). As a change agent, the Agile coach is becoming increasingly critical, serving not just to inspire change and coming up with a methodological target or vision, but taking people along.

The role of a coach in general is at the intersection of intent and action, where they work to close the gap between the ideation of practices and their execution [17]. So, the Agile coaches' idea is to coach a group of people or team towards an agile mindset, most often by adapting behaviors and processes together with them.

Coming back to some methodology aspect that was mentioned before, the Agile coach can leverage AI to enhance the creation and refinement of organizational structures towards an Agile environment (*H-AC2).* However, this requires precise context—again highlighting the importance of skills such as requirements engineering and prompt engineering—to ensure that AI tools are effectively used and come up with the solutions or ideas tailored to the organization's needs (*H-AC2*).

The modern Agile coach must combine traditional coaching techniques, method knowledge on agility with a profound understanding of AI's capabilities. This expertise allows them to guide teams not only in adopting Agile principles and practices but also in integrating AI into their workflows. By doing so, they enable other roles—such as the PO, developer, and SM—to adapt to their evolving responsibilities in an AI-augmented workplace.

In sum, the Agile coach is key to helping teams navigate the complexities of change, encouraging a culture of continuous improvement, and ensuring that the human aspects of Agile are amplified in the face of technological advancements. They act as a bridge between human intuition and AI's analytical prowess, fostering an environment where both can thrive in synergy.

4 | Discussion

The integration of AI into Agile roles represents a paradigm shift in project management and software development—as in many other domains as well. Our examination of the transformed tasks of POs, developers, and SMs has brought up several hypotheses to the mentioned roles working with generative AI (see Table 1).

Besides these role-specific hypothesis, by highlighting the potential of AI to enhance efficiency and innovation, we need to present challenges that necessitate a reevaluation of traditional Agile practices.

4.1 | Synergy and Redefinition of Roles

AI's role in Agile practices is not merely functional but transformative, fostering a deeper synergy between technological possibilities and human expertise. POs are now able to transcend the operational focus on backlog management, leveraging AI to contribute more strategically and creatively to product vision. Developers are shifting from routine coding to complex problem-solving, with AI automating the mundane and freeing human intelligence for higher-order functions. For SMs, AI offers support in administrative tasks, allowing them to focus on the coaching and facilitative aspects of their role, thus becoming more integral in guiding Agile transformations.

4.2 | Ethical and Practical Considerations

Although AI's potential to assume certain tasks within Agile roles is evident, ethical and practical considerations emerge. Dependence on AI for tasks such as backlog management and requirements engineering raises questions about the loss of human judgment in critical decision-making processes [10, 18]. There is a need for a guideline or framework that ensures AI tools are used responsibly—similar to [19] and that the core Agile values of individual and interactions over processes and tools are maintained.

4.3 | The Future of Agile Coaching

The evolving role of the Agile Coach is particularly noteworthy. As the purveyor of the Agile mindset, the Agile Coach's role becomes even more crucial in an AI-driven environment. They must now navigate the intricacies of human-AI collaboration,

TABLE 1 | Summary of all hypothesis.

ID and role	Hypothesis
H-PO1	The product owner focuses on creative and strategic activities instead of management.
H-PO2	The product owner guides GenAI tools by providing the correct and relevant information.
H-DEV1	The developers increase the automation of routine tasks with GenAI tools.
H-DEV2	The developers strongly collaborate with the PO to create high-quality input for AI tools—also with the usage of AI tools.
H-DEV3	Each developer becomes a (software) architect of complex software solutions.
H-SM1	The scrum master increases the automation of administrative tasks with GenAI tools.
H-SM2	The scrum master focuses on practical coaching and increases teaching and training with AI tools.
H-SM3	The scrum master focuses on fostering an agile culture with continuous improvement
H-AC1	The Agile coach focuses on development of soft skills and interpersonal dynamics.
H-AC2	The Agile coach fosters the usage of AI tools and ensures an effective usage.

ensuring that AI tools are employed in ways that enhance rather than undermine human and team dynamics.

5 | Conclusions

This position paper has embarked on a comprehensive exploration of the transformative influence of AI on Agile roles. We have observed the paradigm shift from routine, task-oriented functions towards strategic, creative, and complex problem-solving activities. The PO is evolving into a visionary strategist, the developer into an innovative problem solver, and the SM into a facilitative leader, each enabled by AI's capacity. This change or reshaping of the focus of the different agile roles is manifested in 10 hypotheses of this position paper (see Section 4) that could and should guide us in future research to be validated or not.

The insights garnered point to a future where Agile methodologies and AI converge to create a dynamic, efficient, and human-centered approach to project management. However, this convergence does not come without challenges. It necessitates a vigilant reassessment of Agile's core values and principles, ensuring they remain at the forefront amidst the drive for technological efficiency.

As Agile roles adapt to incorporate AI, there is an imperative need for the Agile community to foster an environment where technology acts as a complement to human expertise, not a substitute. This balance is crucial in maintaining the essence of Agile—its emphasis on people, interactions, and customer collaboration. The Agile community must continue to embrace change, as it always has, but with a renewed focus on ensuring that the technology serves to augment the human elements that are the cornerstone of Agile success.

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Data Availability Statement

Data sharing not applicazble to this article as no datasets were generated or analysed during the current study.

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